

# Supplementary Material for "TFCDiff: Robust ECG Denoising via Time-Frequency Complementary Diffusion"

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## I. SUPPLEMENTARY NOTES

### *Supplementary Note 1 | Comparison of DCT against FFT and DWT*

The Discrete Cosine Transform (DCT) can be viewed as a specialized form of the Discrete Fourier Transform (DFT) that applies even-symmetry extension to the input signal. This preprocessing step confers several critical advantages over the Fast Fourier Transform (FFT) for deep learning applications. Primarily, the real-valued even-symmetry property ensures that the transformation yields purely real coefficients, thereby obviating the need for complex arithmetic. In contrast, FFT generates complex-valued coefficients, which significantly complicates neural network architecture design. Common strategies to handle complex numbers, such as processing real and imaginary components through independent modules or operating solely on magnitudes, invariably increase computational overhead and obscure the modeling of amplitude-phase dependencies. Furthermore, current deep learning frameworks like PyTorch offer limited GPU acceleration for complex operations, posing engineering challenges. Beyond computational efficiency, DCT exhibits superior energy compaction, allowing for more effective signal representation with fewer coefficients. Our preliminary experiments adapting the architecture from [31] for FFT-based inputs confirmed these theoretical drawbacks. The one-generation model incurred prohibitive latency and demonstrated poor denoising performance (SSD = 505.521  $\mu\text{V}^2$ , MAD = 0.790 mV, PRD = 109.173 %, CosSim = 0.838, ImSNR = 0.786 dB), leading us to discard the FFT approach in favor of DCT.

While the Discrete Wavelet Transform (DWT) offers competitive time-frequency localization through Multi-Resolution Analysis (MRA), it presents specific limitations compared to DCT in the context of ECG denoising. First, the performance of DWT is highly sensitive to the selection of the mother wavelet, making it difficult to identify an optimal basis for ECG signals with heavy noise. Second, the strict dyadic downsampling required by DWT imposes rigid constraints on input signal lengths and reduces the flexibility of coefficient truncation. Specifically, input lengths that are not powers of two often cause compatibility errors within U-Net architectures due to mismatched feature map dimensions. To validate this, we replaced the DCT module with a DWT module using the Daubechies-4 wavelet, applying a 2-level decomposition and retaining only the 0 - 45 Hz approximation coefficients. The resulting one-generation model exhibited inferior denoising quality (SSD = 40.232  $\pm$  116.930  $\mu\text{V}^2$ , MAD = 0.521  $\pm$  0.592 mV, PRD = 39.438  $\pm$  27.348 %, CosSim = 0.914  $\pm$  0.110, ImSNR = 11.462  $\pm$  4.570 dB) and slower inference speeds (82.80 ms vs. 43.72 ms per 10-second segment) compared to the DCT-based baseline. Complementing these statistical findings, we conducted time-frequency domain visualizations via Short-Time Fourier Transform (STFT) spectrograms (Fig. S1) to qualitatively assess signal characteristics. Generated using a Hann window (window length and FFT size of 64 samples; hop length of 16 samples), these spectrograms reveal that while DWT yields a globally smoother background, it inadvertently introduces spurious ECG-like waveforms. Such artifacts constitute a critical deficiency for clinical diagnosis. These empirical results, combined with the structural inflexibility of DWT and the risk of generating diagnostic false positives, substantiate our choice of DCT as the most robust and efficient time-frequency representation for this study.

### *Supplementary Note 2 | Validation of Signal Fidelity Under Truncation*

To verify that truncation does not compromise signal fidelity, we evaluated 37,590 ten-second ECG segments from the QT Database (detailed in Section IV-B). Each segment is transformed via 1D DCT, truncated to retain only coefficients corresponding to frequencies below 50 Hz, zero-padded back to the original length, and reconstructed using 1D IDCT. Quantitative comparison between the original and reconstructed signals yielded an average SSD of  $1.68 \times 10^{-3} \mu\text{V}^2$  and an average MAD of  $9.01 \times 10^{-3}$  mV, indicating negligible waveform distortion. Furthermore, the retained low-frequency coefficients accounted for an average of 99.9991 % of the total signal energy. These results conclusively demonstrate that the truncation process preserves essential signal morphology and energy with high precision, validating the signal integrity under truncation.

### Supplementary Note 3 | Forward and Reverse Diffusion Process

We are given a pair of DCT coefficient vectors  $\{\mathbf{d}_0, \tilde{\mathbf{d}}\}$ , where  $\mathbf{d}_0$  denotes the clean coefficients and  $\tilde{\mathbf{d}}$  represents their noisy counterparts, both drawn from an unknown conditional data distribution. Our goal is to learn a model that approximates this conditional distribution, thereby mapping a noisy input  $\mathbf{d}_{noisy}$  to its corresponding noise-free output  $\mathbf{d}_{clean}$ . To this end, we formulate a discrete time conditional diffusion model.

For a discrete timestep  $t \in \{1, 2, \dots, T\}$ , the *forward diffusion process*  $q$  perturbs the clean coefficients  $\mathbf{d}_0$  according to:

$$\mathbf{d}_t = \sqrt{\bar{\alpha}_t} \mathbf{d}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad (1)$$

where  $\mathbf{d}_t$  is the latent representation at step  $t$ ,  $\bar{\alpha}_t \in (0, 1]$  is determined by a predefined noise schedule (detailed in Section III-C), and  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .

When  $t = T$ ,  $x_t$  approximates isotropic Gaussian noise. The forward perturbation kernel  $q(\mathbf{d}_t|\mathbf{d}_0)$  is Gaussian:

$$q(\mathbf{d}_t|\mathbf{d}_0) = \mathcal{N}(\mathbf{d}_t; \sqrt{\bar{\alpha}_t} \mathbf{d}_0, \sqrt{1 - \bar{\alpha}_t} \mathbf{I}). \quad (2)$$

In the *reverse process*  $p$ , we iteratively recover the clean coefficients by denoising  $\mathbf{d}_T$  using a parameterized model  $p(\mathbf{d}_{t-1}|\mathbf{d}_t, \tilde{\mathbf{d}})$ , where the observed noisy coefficients  $\tilde{\mathbf{d}}$  serves as a conditioning input. This reverse process is also modeled as a Gaussian:

$$p(\mathbf{d}_{t-1}|\mathbf{d}_t, \tilde{\mathbf{d}}) = \mathcal{N}(\mathbf{d}_{t-1}; \mu_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}), \sigma_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}) \cdot \mathbf{I}), \quad (3)$$

with mean and standard deviation defined as:

$$\mu_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}) = \frac{1}{\sqrt{\bar{\alpha}_t}} \left( \mathbf{d}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}) \right), \quad (4)$$

$$\sigma_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}) = \begin{cases} \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t, & \text{if } t \geq 1 \\ \beta_1, & \text{if } t = 1 \end{cases}, \quad (5)$$

where  $\epsilon_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}})$  is a U-Net-based noise predictor that predicts the noise component  $\epsilon$ .

Importantly, all inputs in this framework are truncated DCT coefficients as described in Section III-A. After denoising, the output coefficients are zero-padded to their original length and transformed back to the time domain via the 1D IDCT to yield the final denoised ECG signal.

### Supplementary Note 4 | Definitions of Evaluation Metrics

SSD, MAD, PRD, CosSim and ImSNR are defined as follows, with  $h_0$  denoting the ground truth signal,  $\tilde{\mathbf{l}}$  and  $\hat{\mathbf{l}}$  representing the noisy and denoised signals respectively, and  $N$  indicating the signal length.

$$\text{SSD}(\mathbf{l}_0, \hat{\mathbf{l}}) = \sum_{i=1}^N [\mathbf{l}_0(i) - \hat{\mathbf{l}}(i)]^2, \quad (6)$$

$$\text{MAD}(\mathbf{l}_0, \hat{\mathbf{l}}) = \max |\mathbf{l}_0(i) - \hat{\mathbf{l}}(i)|, \quad i \in \{1, 2, \dots, N\}, \quad (7)$$

$$\text{PRD}(\mathbf{l}_0, \hat{\mathbf{l}}) = \sqrt{\frac{\sum_{i=1}^N [\mathbf{l}_0(i) - \hat{\mathbf{l}}(i)]^2}{\sum_{i=1}^N [\hat{\mathbf{l}}(i) - \frac{1}{N} \sum_{i=1}^N \mathbf{l}_0(i)]^2}}, \quad (8)$$

$$\text{CosSim}(\mathbf{l}_0, \hat{\mathbf{l}}) = \frac{\langle \mathbf{l}_0, \hat{\mathbf{l}} \rangle}{\|\mathbf{l}_0\|_2 \|\hat{\mathbf{l}}\|_2}, \quad (9)$$

$$\begin{aligned} \text{ImSNR}(\hat{\mathbf{l}}, \tilde{\mathbf{l}}) &= \text{SNR}(\mathbf{l}_0, \hat{\mathbf{l}}) - \text{SNR}(\mathbf{l}_0, \tilde{\mathbf{l}}) \\ &= 10 \log_{10} \left( \frac{\sum_{i=1}^N [\mathbf{l}_0(i) - \tilde{\mathbf{l}}(i)]^2}{\sum_{i=1}^N [\mathbf{l}_0(i) - \hat{\mathbf{l}}(i)]^2} \right). \end{aligned} \quad (10)$$

### Supplementary Note 5 | Definitions of Fiducial Point Detection Metrics

The performance indicators of fiducial point detection are F1-score (F1) and Mean Absolute Error (MAE), which are defined as follows. Let  $P_{\text{gt}} = \{p_1, p_2, \dots, p_M\}$  denote the set of ground truth fiducial point positions detected in the clean signal, and  $P_{\text{pred}} = \{\hat{p}_1, \hat{p}_2, \dots, \hat{p}_K\}$  denote the set of predicted fiducial point positions in the denoised signal.

To account for physiological tolerance and minor temporal shifts, a predicted fiducial point  $\hat{p}_k$  is considered a True Positive (TP) if it falls within a fixed tolerance window of  $\delta = 200$  ms around an unmatched ground truth fiducial point  $p_m$ . Formally, a match exists if:

$$|\hat{p}_k - p_m| \leq \delta, \quad (11)$$

where each ground truth fiducial point is matched to at most one prediction. Predictions without a corresponding ground truth fiducial point within  $\delta$  are counted as False Positives (FP), while ground truth fiducial points without a matching prediction are counted as False Negatives (FN).

Based on these counts, F1-score for a specific wave type is calculated as the harmonic mean of Precision and Recall:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (12)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (13)$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (14)$$

MAE quantifies the temporal precision of the detected fiducial points while penalizing missed detections to reflect clinical utility. Unlike standard MAE calculations that only average errors over matched pairs, our metric incorporates a penalty for FN based on the average cardiac cycle length. This penalty mechanism prevents models that tend to predict fiducial points only on simple samples from achieving artificially lower MAE scores. Specifically, it is computed as:

$$\text{MAE} = \frac{1}{N_{\text{gt}}} \left( \sum_{i \in \mathcal{M}} |p_i^{(\text{gt})} - \hat{p}_i^{(\text{pred})}| + \text{FN} \cdot \overline{RR} \right), \quad (15)$$

where  $N_{\text{gt}}$  denotes the total number of ground truth fiducial points,  $\mathcal{M}$  denotes the set of indices for successfully matched pairs,  $p_i^{(\text{gt})}$  and  $\hat{p}_i^{(\text{pred})}$  represent the time indices of the  $i$ -th matched ground truth and predicted points, respectively, and  $\overline{RR}$  is the average RR interval of the signal. A lower MAE value indicates both high detection completeness and precise temporal alignment.

In our experiments, both metrics are computed separately for P, Q, R, S, and T waves, and the final reported values represent the macro-average across all test samples in the synthesized dataset.

## II. SUPPLEMENTARY FIGURES

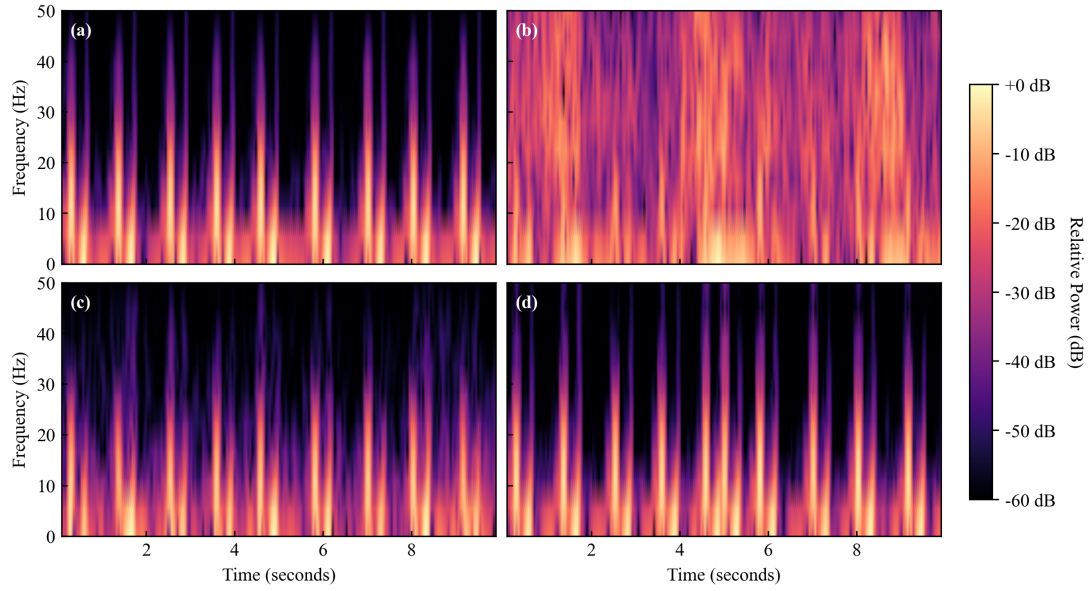


Fig. S1. Time-frequency domain analysis via STFT spectrograms comparing the denoising performance of applying DCT and DWT. (a) Clean ground truth. (b) Noisy input. (c) The signal denoised by TFCDiff with DCT. (d) The signal denoised by TFCDiff with DWT.

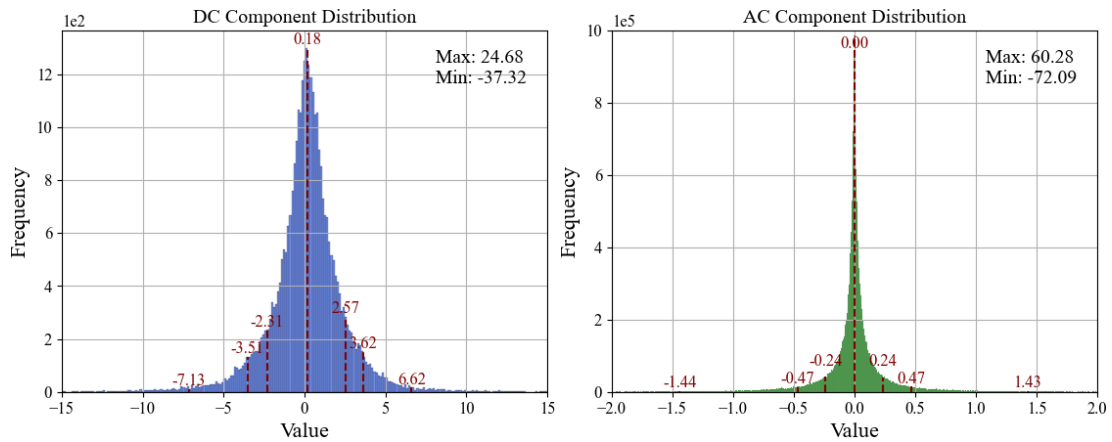


Fig. S2. Distributions of the DC and AC components. Maroon lines indicate the 1st, 5th, 10th, 50th, 90th, 95th, and 99th percentiles for both components.



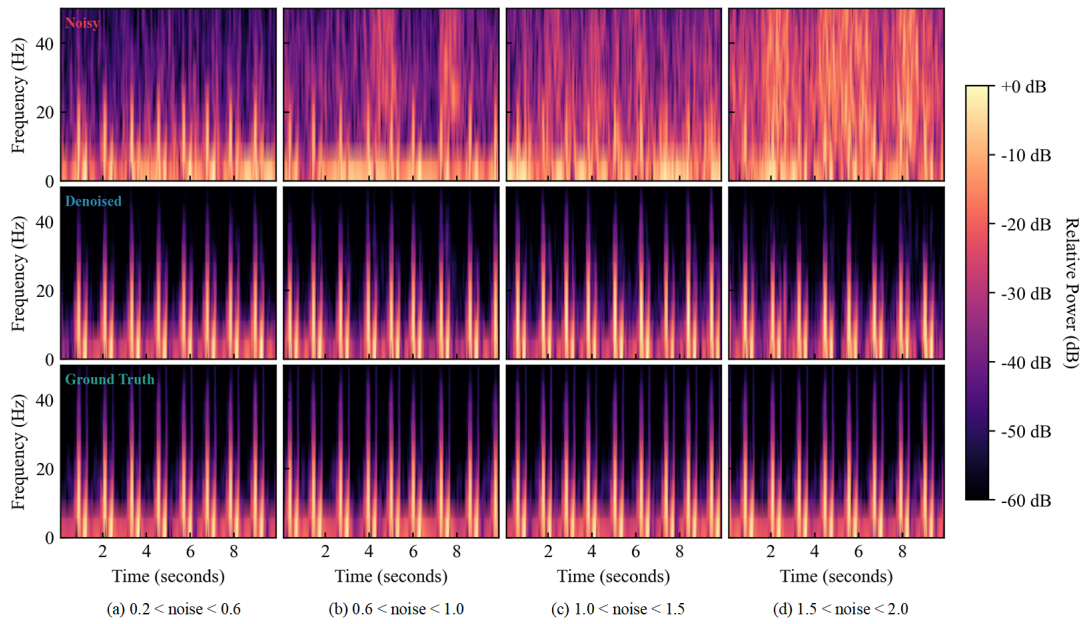


Fig. S3. Time-frequency domain analysis of TFCDiff denoised results on the synthesized dataset via STFT spectrograms generated using a Hann window (window length and FFT size of 64 samples, hop length of 16 samples). (a) Noise Level: 0.2–0.6. (b) Noise Level: 0.6–1.0. (c) Noise Level: 1.0–1.5. (d) Noise Level: 1.5–2.0.

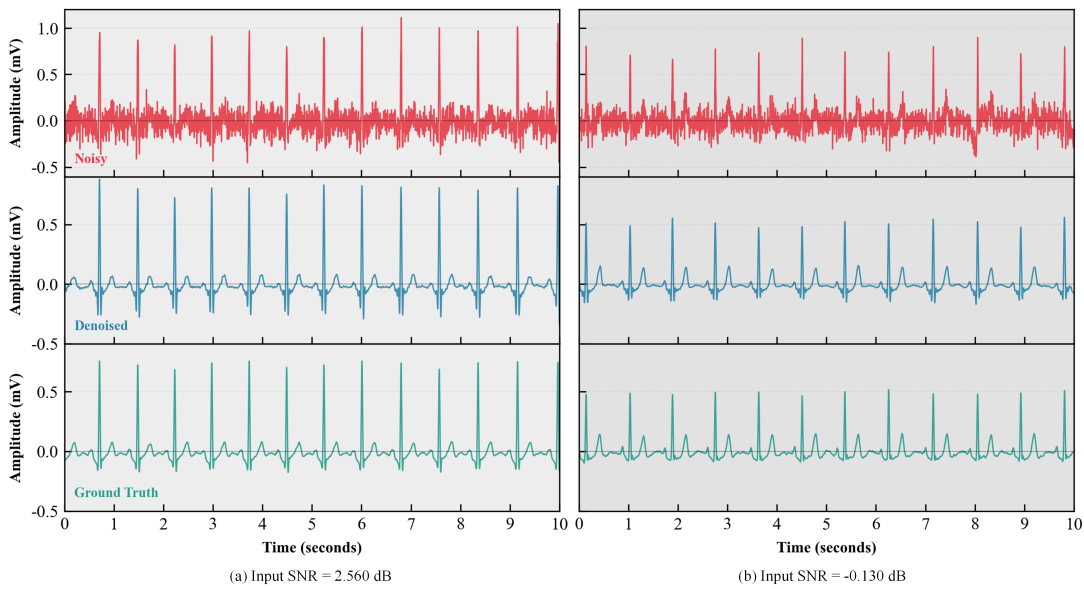


Fig. S4. Visualization of TFCDiff denoised results on the SimEMG Database with representative samples. (a) Input SNR 2.560 dB: SSD 2.644  $\mu\text{V}^2$ , MAD 0.175 mV, PRD 19.258 %, CosSim 0.987, ImSNR 10.681 dB. (b) Input SNR -0.130 dB: SSD 1.684  $\mu\text{V}^2$ , MAD 0.137 mV, PRD 22.995 %, CosSim 0.975, ImSNR 12.208 dB.

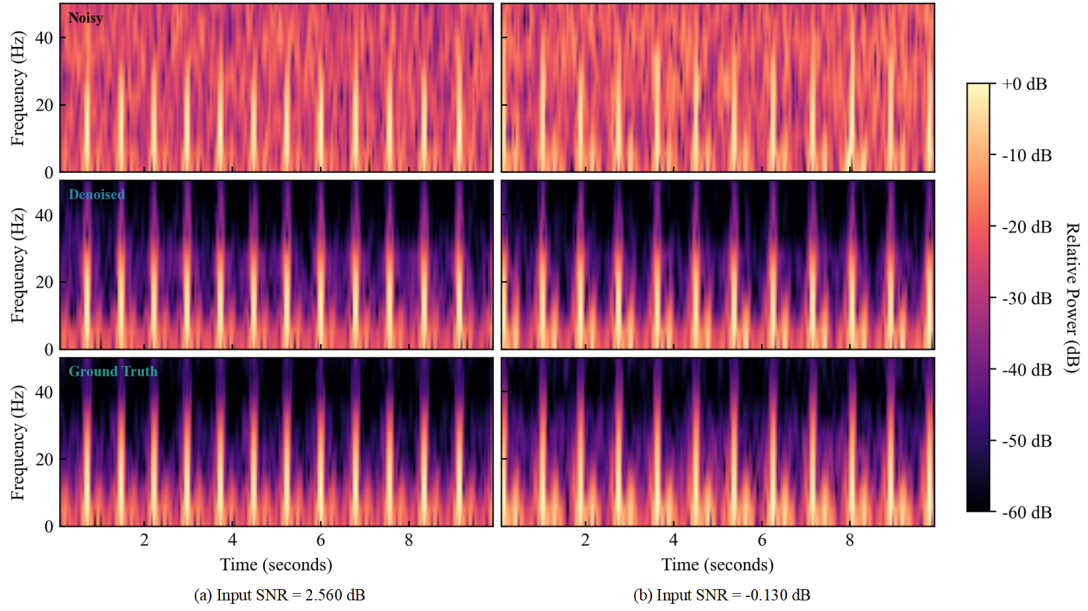


Fig. S5. Time-frequency domain analysis of TFCDiff denoised results on the SimEMG Database via STFT spectrograms generated using a Hann window (window length and FFT size of 64 samples, hop length of 16 samples). (a) Input SNR: 2.560 dB. (b) Input SNR: -0.130 dB.

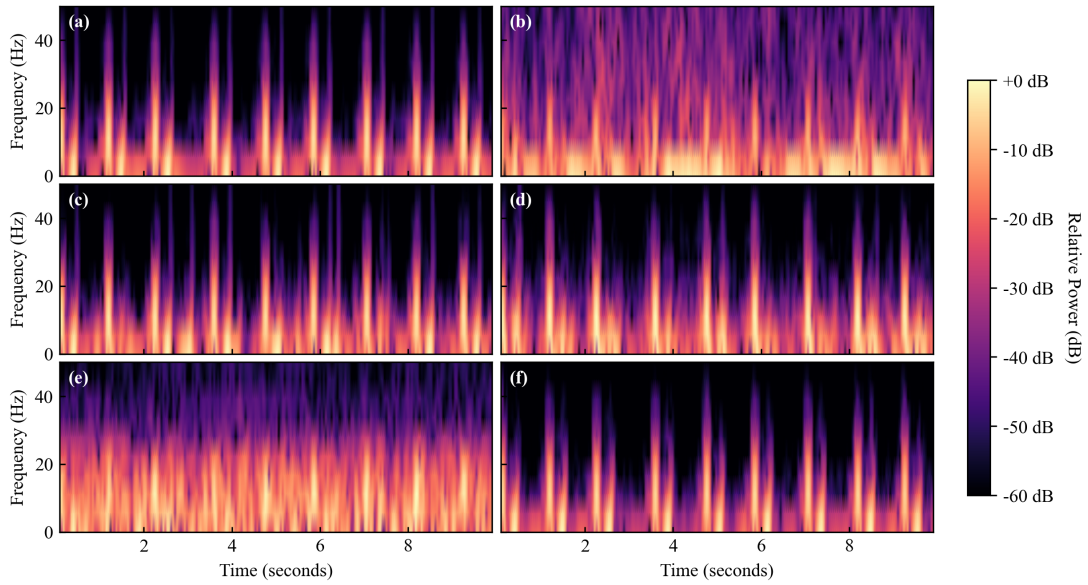


Fig. S6. Time-frequency domain analysis of denoised results in the ablation study of DCT diffusion and TFEM via STFT spectrograms generated using a Hann window (window length and FFT size of 64 samples, hop length of 16 samples). (a) Ground truth signal. (b) Noisy ECG signal. (c) Time-domain diffusion only. (d) Time-domain diffusion with TFEM. (e) DCT diffusion only. (f) DCT diffusion with TFEM.

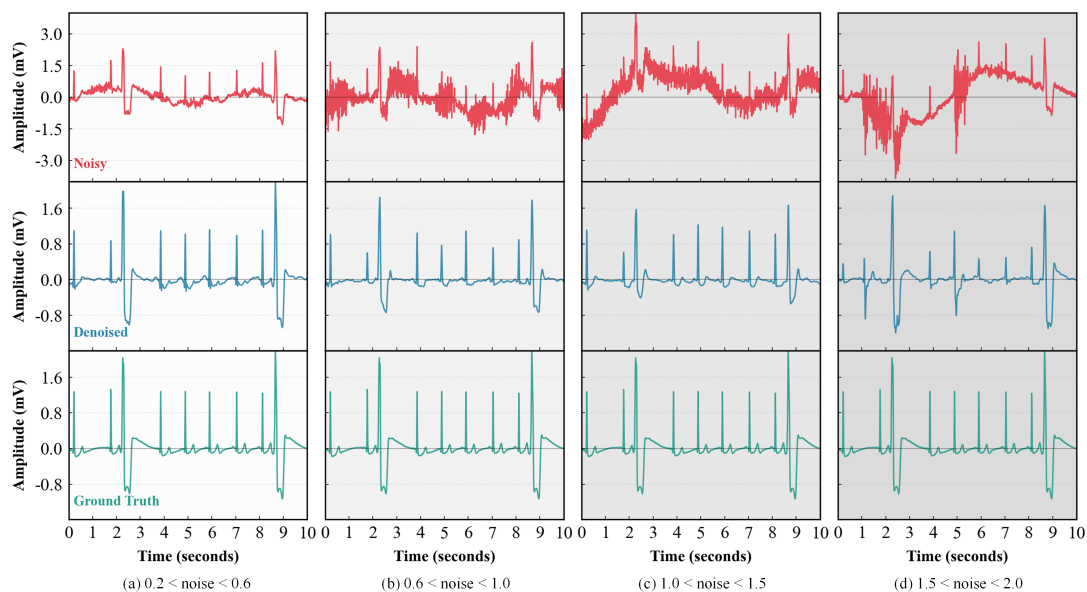


Fig. S7. Visualization of TFCDiff denoised results on PVC signals across varying noise intensities.

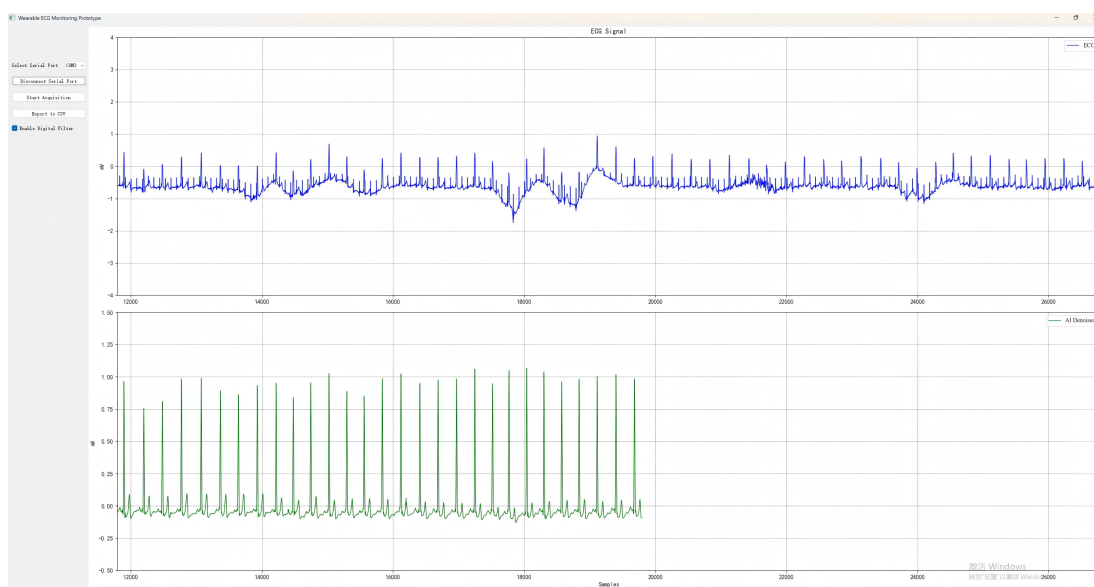


Fig. S8. Prototype of the developed wearable ECG monitoring system, showcasing the host software interface for real-time data acquisition and synchronized denoising.

## III. SUPPLEMENTARY TABLES

TABLE S1  
SUMMARY OF DATASETS

| Dataset         | QTDB | NSTDB | SimEMG |
|-----------------|------|-------|--------|
| Recordings      | 105  | 3     | 220    |
| Length (second) | 900  | 1800  | 30     |
| Channel         | 2    | 2     | 1      |
| Fs (Hz)         | 250  | 360   | 500    |

TABLE S2  
RECORDINGS SELECTED FOR TEST SET

| Database                                     | Recording ID       |
|----------------------------------------------|--------------------|
| MIT-BIH Arrhythmia Database                  | sel123, sel233     |
| MIT-BIH ST Change Database                   | sel302, sel307     |
| MIT-BIH Supraventricular Arrhythmia Database | sel820, sel853     |
| MIT-BIH Normal Sinus Rhythm Database         | sel16420, sel16795 |
| European ST-T Database                       | sel0106, sel0121   |
| Sudden death patients from BIH               | sel32, sel49       |
| MIT-BIH LongTerm ECG Database                | sel14046, sel15814 |

TABLE S3  
SSD ( $\mu V^2$ ) COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

| Models            | 0.2-0.6             | 0.6-1.0              | 1.0-1.5              | 1.5-2.0              |
|-------------------|---------------------|----------------------|----------------------|----------------------|
| FIR               | 17.414±38.569       | 53.222±99.580        | 125.252±260.503      | 233.005±465.136      |
| IIR               | 23.732±62.589       | 58.063±108.847       | 128.126±259.979      | 234.929±463.763      |
| DRNN              | 42.077±97.849       | 55.626±125.557       | 75.556±171.716       | 98.773±208.981       |
| FCN-DAE           | 28.308±49.248       | 32.132±54.977        | 39.145±72.682        | 50.431±86.602        |
| Multibranch LANLD | 21.255±36.343       | 29.227±47.199        | 39.379±69.582        | 53.981±90.000        |
| DesCod-1          | 16.599±40.581       | 24.596±61.984        | 36.821±93.622        | 52.919±116.625       |
| DesCod-3          | 12.253±39.038       | 18.544±57.313        | 27.147±73.191        | 38.675±86.161        |
| DesCod-5          | 11.114±34.247       | 17.600±53.325        | 25.457±71.325        | 36.858±88.830        |
| DesCod-10         | 10.708±36.446       | 16.266±51.675        | 23.838±66.124        | 35.127±84.742        |
| TCDAE             | 10.703±23.814       | 15.012±32.874        | 22.207±52.657        | 33.138±72.822        |
| TFCDiff-1         | 13.215±49.785       | 20.622±63.198        | 30.499±89.642        | 42.378±107.589       |
| TFCDiff-3         | 10.791±44.262       | 15.421±48.989        | 23.594±76.668        | 33.245±89.130        |
| TFCDiff-5         | 10.153±42.118       | 14.676±48.814        | 22.430±72.247        | 31.061±83.895        |
| TFCDiff-10        | <b>9.677±40.306</b> | <b>14.222±46.310</b> | <b>21.512±70.509</b> | <b>29.497±79.668</b> |

TABLE S4  
MAD (mV) COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

| Models            | 0.2-0.6            | 0.6-1.0            | 1.0-1.5            | 1.5-2.0            |
|-------------------|--------------------|--------------------|--------------------|--------------------|
| FIR               | 0.242±0.223        | 0.440±0.346        | 0.665±0.536        | 0.947±0.763        |
| IIR               | 0.332±0.439        | 0.499±0.417        | 0.701±0.580        | 0.971±0.778        |
| DRNN              | 0.425±0.466        | 0.517±0.513        | 0.641±0.585        | 0.775±0.599        |
| FCN-DAE           | 0.481±0.473        | 0.488±0.428        | 0.542±0.488        | 0.603±0.464        |
| Multibranch LANLD | 0.280±0.293        | 0.347±0.294        | 0.411±0.330        | 0.499±0.368        |
| DesCod-1          | 1.056±1.095        | 1.045±1.071        | 1.095±1.054        | 1.212±1.075        |
| DesCod-3          | 0.686±0.589        | 0.719±0.610        | 0.763±0.597        | 0.834±0.606        |
| DesCod-5          | 0.580±0.481        | 0.611±0.488        | 0.650±0.489        | 0.727±0.521        |
| DesCod-10         | 0.458±0.394        | 0.490±0.403        | 0.517±0.416        | 0.618±0.474        |
| TCDAE             | 0.216±0.266        | 0.258±0.253        | 0.337±0.334        | 0.448±0.399        |
| TFCDiff-1         | 0.214±0.377        | 0.287±0.360        | 0.376±0.434        | 0.490±0.474        |
| TFCDiff-3         | 0.193±0.368        | 0.255±0.320        | 0.332±0.396        | 0.429±0.437        |
| TFCDiff-5         | 0.190±0.361        | 0.249±0.318        | 0.323±0.387        | 0.416±0.417        |
| TFCDiff-10        | <b>0.186±0.347</b> | <b>0.244±0.319</b> | <b>0.316±0.381</b> | <b>0.404±0.406</b> |

TABLE S5  
PRD (%) COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

| Models            | 0.2-0.6              | 0.6-1.0              | 1.0-1.5              | 1.5-2.0              |
|-------------------|----------------------|----------------------|----------------------|----------------------|
| FIR               | 26.222±8.911         | 42.957±10.788        | 57.398±12.384        | 68.943±11.535        |
| IIR               | 29.103±10.469        | 44.839±10.996        | 58.331±12.181        | 69.591±11.374        |
| DRNN              | 54.682±26.012        | 63.227±26.219        | 72.672±25.243        | 82.869±22.559        |
| FCN-DAE           | 44.133±32.204        | 45.370±25.707        | 52.251±29.742        | 58.609±25.216        |
| Multibranch LANLD | 37.009±16.882        | 45.729±18.433        | 55.301±21.659        | 63.835±19.963        |
| DesCod-1          | 27.457±14.801        | 33.635±16.180        | 42.735±21.211        | 52.751±23.157        |
| DesCod-3          | 22.428±11.772        | 28.613±14.647        | 37.266±20.029        | 46.841±22.564        |
| DesCod-5          | 20.896±10.368        | 27.414±14.705        | 35.719±19.796        | 45.277±23.174        |
| DesCod-10         | 19.553±10.260        | 25.796±13.824        | 34.238±19.345        | 44.142±23.278        |
| TCDAE             | 21.258±9.321         | 25.393±9.537         | 31.389±12.939        | 38.027±15.648        |
| TFCDiff-1         | 19.003±11.936        | 26.624±15.408        | 34.451±20.583        | 43.463±25.293        |
| TFCDiff-3         | 16.310±11.206        | 22.798±13.401        | 30.054±19.061        | 38.986±25.401        |
| TFCDiff-5         | 15.645±11.059        | 21.951±12.547        | 29.166±19.576        | 37.790±24.721        |
| TFCDiff-10        | <b>15.179±10.830</b> | <b>21.358±12.943</b> | <b>28.390±19.596</b> | <b>36.726±24.850</b> |

TABLE S6  
COSSIM COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

| Models            | 0.2-0.6            | 0.6-1.0            | 1.0-1.5            | 1.5-2.0            |
|-------------------|--------------------|--------------------|--------------------|--------------------|
| FIR               | 0.961±0.028        | 0.895±0.056        | 0.805±0.091        | 0.708±0.108        |
| IIR               | 0.951±0.041        | 0.886±0.059        | 0.799±0.091        | 0.702±0.109        |
| DRNN              | 0.922±0.037        | 0.884±0.057        | 0.825±0.087        | 0.749±0.111        |
| FCN-DAE           | 0.919±0.090        | 0.913±0.076        | 0.886±0.098        | 0.853±0.104        |
| Multibranch LANLD | 0.950±0.026        | 0.924±0.038        | 0.887±0.063        | 0.843±0.079        |
| DesCod-1          | 0.953±0.067        | 0.938±0.068        | 0.903±0.095        | 0.860±0.115        |
| DesCod-3          | 0.972±0.035        | 0.958±0.043        | 0.933±0.066        | 0.901±0.086        |
| DesCod-5          | 0.977±0.024        | 0.963±0.037        | 0.941±0.056        | 0.911±0.079        |
| DesCod-10         | 0.980±0.022        | 0.968±0.031        | 0.947±0.052        | 0.918±0.072        |
| TCDAE             | 0.977±0.024        | 0.967±0.028        | 0.949±0.047        | 0.924±0.068        |
| TFCDiff-1         | 0.980±0.030        | 0.963±0.044        | 0.939±0.069        | 0.906±0.102        |
| TFCDiff-3         | 0.985±0.025        | 0.974±0.029        | 0.956±0.052        | 0.930±0.080        |
| TFCDiff-5         | 0.986±0.023        | 0.976±0.026        | 0.959±0.050        | 0.937±0.071        |
| TFCDiff-10        | <b>0.987±0.022</b> | <b>0.977±0.025</b> | <b>0.962±0.048</b> | <b>0.941±0.067</b> |

TABLE S7  
IMSNR (dB) COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

| Models            | 0.2-0.6             | 0.6-1.0             | 1.0-1.5             | 1.5-2.0             |
|-------------------|---------------------|---------------------|---------------------|---------------------|
| FIR               | 5.265±3.260         | 6.388±3.281         | 6.825±3.502         | 6.890±3.400         |
| IIR               | 4.375±3.726         | 5.987±3.296         | 6.676±3.519         | 6.816±3.397         |
| DRNN              | 1.441±3.925         | 6.387±3.176         | 8.898±2.876         | 10.316±2.553        |
| FCN-DAE           | 2.388±4.042         | 8.110±3.311         | 11.062±3.125        | 12.758±2.921        |
| Multibranch LANLD | 3.597±3.906         | 8.156±3.315         | 10.671±3.216        | 12.191±2.842        |
| DesCod-1          | 5.630±5.092         | 10.234±4.164        | 12.299±4.019        | 13.307±3.716        |
| DesCod-3          | 7.464±4.521         | 11.832±3.938        | 13.888±3.905        | 14.907±3.686        |
| DesCod-5          | 8.067±4.339         | 12.289±3.883        | 14.376±3.808        | 15.363±3.678        |
| DesCod-10         | 8.690±4.230         | 12.846±3.811        | 14.852±3.793        | 15.722±3.721        |
| TCDAE             | 7.706±4.385         | 12.127±3.618        | 14.300±3.466        | 15.549±3.301        |
| TFCDiff-1         | 8.995±3.881         | 12.410±3.514        | 14.333±3.478        | 15.287±3.516        |
| TFCDiff-3         | 10.504±3.901        | 13.922±3.679        | 15.779±3.547        | 16.716±3.670        |
| TFCDiff-5         | 10.905±3.939        | 14.276±3.695        | 16.139±3.597        | 17.094±3.704        |
| TFCDiff-10        | <b>11.217±3.983</b> | <b>14.601±3.741</b> | <b>16.454±3.658</b> | <b>17.412±3.765</b> |